The AI Infrastructure Gap: Why Big Tech's 730 Billion Dollar Bet Doesn't Add Up Yet

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Introduction

On June 25, Apple raised prices on its Macs and iPads in the middle of a product cycle, saying it could no longer absorb the cost of memory chips. The reason traces back to AI: the companies building AI data centers are buying so much memory that everyday electronics now cost more.

That is AI's strange position in late 2026. It is technically remarkable: it writes code, reads contracts and carries out multi-step tasks that a few years ago needed a person. But I study finance, and on the financial side a large part of the picture does not add up yet. Amazon, Alphabet, Microsoft and Meta are on track to spend roughly $730 billion on capital projects in 2026, mostly AI data centers, chips and power. The two biggest AI labs, OpenAI and Anthropic, bring in about $115 billion a year in revenue between them. This paper follows that gap: how the spending got so large, why it continues, what the gap means, and how memory chips are carrying the cost into consumers' hands.

Spending Out of Fear

The build-out has roughly tripled in two years. Combined capital spending at Amazon, Alphabet, Microsoft and Meta was about $226 billion in 2024 and $410 billion in 2025. The 2026 plans are: Amazon about $220 billion, Alphabet $195-205 billion, Meta $135-145 billion, and Microsoft about $170 billion. That adds up to roughly $720-740 billion. Against it, Anthropic's run-rate revenue was $65 billion at the end of July and OpenAI's is about $50 billion. A run rate is the current pace projected over a year, and it is revenue, not profit. Even so, it is about $6 of spending for every $1 the labs bring in.

Markets have noticed. On June 5, U.S. chip stocks lost about $1.3 trillion in a day after Broadcom reported that demand for its custom AI chips fell short of lofty expectations. Nvidia fell about 6%, cutting more than $300 billion from its market value, and Micron fell 13%. The selloff did not end the boom, but it showed how much of the boom rests on expectations.

Why keep spending? I read it as fear more than ambition. Think of BlackBerry: it led the early smartphone market, was slow to answer the touchscreen, and lost the market. No big tech CEO wants to be the BlackBerry of AI. They say as much. In 2024 Alphabet's Sundar Pichai told investors that the risk of underinvesting is dramatically greater than the risk of overinvesting. Meta's Mark Zuckerberg admitted companies may be overbuilding, but argued that falling behind on the most important technology of the next decade is worse. Sequoia's David Cahn, who asked in 2024 where the revenue to justify the build-out would come from, called it an arms race driven by game theory and fear of missing out, not by revenue.

The cost is showing up in cash. In the latest quarter, Amazon's trailing twelve-month free cash flow was negative $42.4 billion, Alphabet's was negative $9.07 billion for the quarter and Meta's negative $2.12 billion. Microsoft was the only one still positive, and it said it will now turn to the bond market. One analysis of the biggest spenders found that capex now takes about 95% of operating cash flow, up from about 50% in 2024. In plain terms, most of the profit is going back into data centers, not out to shareholders.

A Factory Bigger Than Its Market

Imagine a company that spends $10 million building a coffee machine factory, then sells $400,000 worth of coffee a year. The machines work and the customers like the coffee, but each $1 of sales needs $25 of factory behind it. No spreadsheet turns that into a good investment.

AI infrastructure is that factory at a gentler ratio, about $6 of spending per $1 of lab revenue, and the factory is still being expanded. Amazon's capex is projected by analysts to top $280 billion next year.

We have seen this movie before. In the late 1990s, telecom companies laid about $90 billion of fiber-optic cable in four years, betting that internet traffic would fill it. By 2001, Merrill Lynch estimated only about 2.6% of that capacity was in use, and more than 60 telecom companies failed between 2001 and 2003. The technology was not wrong. The cable stayed in the ground and later carried streaming and cloud traffic. But the companies that paid for it were mostly not the ones that profited.

Gartner put numbers on the same problem on October 14. AI providers will spend nearly $1 trillion on infrastructure in 2026 and plan about $6 trillion more from 2027 to 2030. To pay for that, they would need roughly $8.5 trillion of cumulative revenue through 2030. Gartner expects about $3.6 trillion at current growth, nearly $5 trillion short.

The finance term for what happens next is impairment: a write-down when an asset is worth less than its cost. Gartner says more than $3.5 trillion of AI infrastructure could be exposed to it by 2030. It sees the most likely fix as spending falling back in line with

demand, and expects over 80% of independent AI-focused cloud providers to stop operating on their own by then.

There is a fair counterpoint. The factory is selling more coffee quickly: Anthropic reported $16.8 billion of third-quarter revenue, and Gartner calls its scenario the most likely one, not the only one. But a gap this large needs demand to grow much faster than it has, or something has to give.

DRAM: How AI Made Your Next Phone More Expensive

DRAM is the working memory inside every phone, laptop and server. Samsung, SK Hynix and Micron make most of it, and AI has changed what they want to make.

AI chips need high-bandwidth memory (HBM), stacks of DRAM placed next to the processor. HBM earns higher margins, and it takes roughly three times the factory capacity per gigabyte that standard DRAM does. A factory's capacity is a fixed pie, so every wafer that becomes HBM is a wafer that does not become memory for a phone. Samsung followed the money: one of its executives said HBM will take almost 30% of the industry's DRAM wafer capacity in 2027, up from about 20%. SK Hynix has said its memory capacity is essentially sold out for 2026, and Micron has exited the consumer memory market to focus on enterprise and AI customers.

Less supply for everyone else means higher prices:

- **Memory.** DRAM contract prices rose as much as 98% in the first quarter, according to TrendForce. It expects another 10-15% in the fourth quarter.
- **Devices.** In June, Apple said it could no longer absorb memory costs. A 512 GB MacBook Air went from $1,099 to $1,299. IDC forecasts average PC prices will rise 17% in 2026.
- **Samsung itself.** Its own marketing chief warned in January that the shortage would affect everyone, not just Samsung.

Relief is not close. Micron's CEO expects supply and demand to be much tighter in 2027 and 2028 than in 2026, and Micron has already committed over 75% of its 2027 output. The builders feel it too: Amazon said higher memory costs pushed its capex up, and Microsoft said about $25 billion of an earlier $190 billion plan was component inflation, not new capacity.

That is the clearest sign of the gap in everyday terms. The economics are strained at the top, and the bill is already landing on people who have never used an AI data center.